Fluid Mechanics ME 21201 Quiz 3

October 30, 2026

1. (1 pt) In fully developed internal pipe flow, what does a high Reynolds number fundamentally signify regarding the balance of forces within the fluid?

   A. Viscous forces are much larger than inertial forces, suppressing rapid velocity fluctuations.
   B. Gravity and surface tension forces are equal, creating capillary waves inside the pipe.
   C. Inertial forces dominate over viscous forces, making the flow susceptible to destabilization and turbulence.
   D. Pressure gradients completely vanish, causing flow velocity to become uniform across the cross-section.

   Inertial forces dominate over viscous forces, making the flow susceptible to destabilization and turbulence.

2. (1 pt) How does the hydrodynamic entrance length (L_h) for a turbulent pipe flow compare to that of a laminar pipe flow at the same Reynolds number?

   A. Turbulent entrance length is much longer because turbulent eddies take more distance to organize into a steady profile.
   B. They are identical because entrance length depends solely on the pipe diameter and fluid density, regardless of flow regime.
   C. Turbulent entrance length is zero because turbulence prevents boundary layer formation entirely.
   D. Turbulent entrance length is significantly shorter than laminar entrance length because turbulent mixing accelerates boundary layer growth and core uniform velocity stabilization.

   Turbulent entrance length is significantly shorter than laminar entrance length because turbulent mixing accelerates boundary layer growth and core uniform velocity stabilization.

3. (1 pt) In the completely rough (fully rough) turbulent zone of the Moody chart, how does the Darcy friction factor (f) behave with respect to the Reynolds number (Re)?

   A. f becomes completely independent of the Reynolds number and depends only on the relative roughness (ϵ/D).
   B. f decreases linearly as Re increases.
   C. f increases proportionally with the square root of Re.
   D. f depends inversely on Re raised to the power of 0.25

   f becomes completely independent of the Reynolds number and depends only on the relative roughness (ϵ/D).

4. (1 pt) Why is the velocity profile in fully developed turbulent pipe flow much flatter (more uniform) near the center compared to a laminar flow profile?

   A. Intense cross-stream momentum mixing by turbulent eddies and vortices rapidly transfers momentum across the flow cross-section.
   B. Molecular diffusion is infinitely high in turbulent flow, eliminating all velocity gradients across the entire pipe radius.
   C. The no-slip condition is violated at high Reynolds numbers, allowing the centerline velocity to stretch outward.
   D. Fluid density drops near the pipe axis in turbulent flow, reducing local inertia effects.

   Intense cross-stream momentum mixing by turbulent eddies and vortices rapidly transfers momentum across the flow cross-section.

5. (2 pts) For fully developed laminar flow in a horizontal circular pipe, frictional head loss (h_f) is proportional to which power of the average flow velocity (V)?

A. Independent
B. Quadratic
C. Square Root
D. Linear

Linear

6. (1 pt) Which of the following statements is conceptually correct regarding minor losses (e.g., in valves, bends, or sudden expansions) in pipe flow?

A. Minor losses are caused exclusively by molecular friction at the pipe walls over extended straight lengths.
B. The loss coefficient K_L for a specific fitting always increases linearly as the fluid velocity increases.
C. Minor losses are primarily caused by flow separation, turbulence generation, and flow recirculation resulting from abrupt geometrical changes.
D. Minor losses are completely negligible in turbulent flow and only matter in laminar regimes.

Minor losses are primarily caused by flow separation, turbulence generation, and flow recirculation resulting from abrupt geometrical changes.

7. (1 pt) When applying the hydraulic diameter ($D_h = 4A/P$) to non-circular ducts for calculating the friction factor in turbulent flow, why is it an approximation rather than an exact solution?

A. The wetted perimeter P changes along the flow direction due to compressibility effects.
B. Cross-sectional area A is no longer defined for non-symmetric shapes.
C. The Reynolds number formula becomes completely invalid and cannot be evaluated.
D. Secondary flows induced in the corners of non-circular ducts alter the shear stress distribution and turbulence structure compared to circular pipes.

Secondary flows induced in the corners of non-circular ducts alter the shear stress distribution and turbulence structure compared to circular pipes.

8. (2 pts) Why is the kinetic energy correction factor (α) equal to 2.0 for fully developed laminar flow in a circular pipe, whereas it is approximately 1.05 to 1.15 for turbulent flow?

A. Laminar flow has a flat velocity profile, making kinetic energy calculations uniform across the cross-section and giving a value of 2.0.
B. Turbulent flow has zero shear stress at the wall, making velocity uniform.
C. Laminar flow features a steep parabolic velocity profile where peak centerline velocity is twice the average velocity.
D. Compressibility effects are massive in laminar pipe flow but negligible in turbulent flow.

Laminar flow features a steep parabolic velocity profile where peak centerline velocity is twice the average velocity, skewing the actual kinetic energy flux compared to using average velocity cubed.